Tutorial 4: Systems of Linear Equations

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1. For each of the following systems of linear equations: (i) express the linear system as a matrix product $\boldsymbol{A}\vec{x} = \vec{b}$; (ii) perform Gaussian elimination to reduce the augmented matrix $(\boldsymbol{A} \mid \boldsymbol{b})$ into row-echelon / reduced row-echelon form, and (iii) determine the solution(s), if any, to the linear system.

(a) $\begin{cases} x + 2y - 4z = -4 \\ 2x + 5y - 9z = -10 \\ 3x - 2y + 3z = 11 \end{cases}$

(b) $\begin{cases} x_1 + 2x_2 - 3x_3 + 4x_4 = 2 \\ 2x_1 + 5x_2 - 2x_3 + x_4 = 1 \\ 5x_1 + 12x_2 - 7x_3 + 6x_4 = 3 \end{cases}$

(c) $\begin{cases} x_1 - 3x_2 + 2x_3 - x_4 + 2x_5 = 2 \\ 3x_1 - 9x_2 + 7x_3 - x_4 + 3x_5 = 7 \\ 2x_1 - 6x_2 + 7x_3 + 4x_4 - 5x_5 = 7 \end{cases}$

(a) We have the matrix product

$$\begin{pmatrix} 1 & 2 & -4 \\ 2 & 5 & -9 \\ 3 & -2 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -4 \\ -10 \\ 11 \end{pmatrix}.$$

Performing Gaussian elimination, we have

$$\left(\begin{array}{ccc|c} 1 & 2 & -4 & -4 \\ 2 & 5 & -9 & -10 \\ 3 & -2 & 3 & 11 \end{array}\right) \xrightarrow[R_3-3R_1]{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -4 & -4 \\ 0 & 1 & -1 & -2 \\ 0 & -8 & 15 & 23 \end{array}\right) \cdots$$

$$\xrightarrow{R_3+8R_2} \left(\begin{array}{ccc|c} 1 & 2 & -4 & -4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 7 & 7 \end{array}\right) \xrightarrow{\frac{1}{7}R_3} \left(\begin{array}{ccc|c} 1 & 2 & -4 & -4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right) \cdots$$

$$\xrightarrow[R_2+R_3]{R_1+4R_3} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_1-2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right).$$

We thus have a unique solution: $x = 2$, $y = -1$, and $z = 1$.

(b) We have the matrix product

$$\begin{pmatrix} 1 & 2 & -3 & 4 \\ 2 & 5 & -2 & 1 \\ 5 & 12 & -7 & 6 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

Performing Gaussian elimination, we have

$$\left(\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 2 \\ 2 & 5 & -2 & 1 & 1 \\ 5 & 12 & -7 & 6 & 3 \end{array}\right) \xrightarrow[R_3-5R_1]{R_2-2R_1} \xrightarrow{R_3-2R_2} \left(\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 2 \\ 0 & 1 & 4 & -7 & -3 \\ 0 & 0 & 0 & 0 & -1 \end{array}\right).$$

The system is inconsistent and has no solution.

(c) We have the matrix product

$$\begin{pmatrix} 1 & -3 & 2 & -1 & 2 \\ 3 & -9 & 7 & -1 & 3 \\ 2 & -6 & 7 & 4 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \\ 7 \end{pmatrix}.$$

Performing Gaussian elimination, we have

$$\left(\begin{array}{ccccc|c} 1 & -3 & 2 & -1 & 2 & 2 \\ 3 & -9 & 7 & -1 & 3 & 7 \\ 2 & -6 & 7 & 4 & -5 & 7 \end{array}\right) \xrightarrow[R_3-2R_1]{R_2-3R_1} \xrightarrow[R_3-3R_2]{R_1-2R_2} \left(\begin{array}{ccccc|c} 1 & -3 & 0 & -5 & 8 & 0 \\ 0 & 0 & 1 & 2 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

The system has infinitely many solutions. Leting $x_2 = r$, $x_4 = s$, and $x_5 = t$, we have

$$x_1 = 3r + 5s - 8t,\ x_2 = r,\ x_3 = 1 - 2s + 3t,\ x_4 = s,\ x_5 = t,\ \text{for } s, t \in \mathbb{R}.$$

2. (a) Consider the following linear system:

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + \left(a^2 - 14\right) z = a + 2 \end{cases}.$$

For what values of a will the system have (i) no solution? (ii) exactly one solution? (iii) infinitely many solutions?

(b) Consider the following linear system:

$$\begin{cases} ax + y = a \\ x + y + z = 1 \\ y + az = b \end{cases}.$$

For what values of a and b will the system have (i) no solution? (ii) exactly one solution? (iii) infinitely many solutions?

(a) Performing Gaussian elimination on the augmented matrix,

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & a^2-14 & a+2 \end{array}\right) \xrightarrow[R_3-4R_1]{R_2-3R_1} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & a^2-2 & a-14 \end{array}\right) \cdots$$

$$\cdots \xrightarrow{R_3-R_2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & a^2-16 & a-4 \end{array}\right).$$

The last row in the above matrix corresponds to the equation

$$(a+4)(a-4)z = a-4.$$

- There are infinitely many solutions if we have a row corresponding to $0 = 0$. Hence, there are infinitely many solutions if $a = 4$.
- There are no solutions when $0 = c$ $(c \neq 0)$; i.e., when $a = -4$.
- Otherwise, there is a unique solution, i.e., when $a \in \mathbb{R} \setminus \{-4, 4\}$.

(b) We row reduce the augmented matrix:

$$\left(\begin{array}{ccc|c} a & 1 & 0 & a \\ 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ a & 1 & 0 & a \\ 0 & 1 & a & b \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \\ a & 1 & 0 & a \end{array}\right) \cdots$$

$$\cdots \xrightarrow{R_3-aR_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \\ 0 & 1-a & -a & 0 \end{array}\right) \xrightarrow{R_3-(1-a)R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \\ 0 & 0 & a(a-2) & -b(1-a) \end{array}\right).$$

When $a(a-2) \neq 0$, then the system has a unique solution. When $a(a-2) = 0$, the final augmented matrix becomes

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \\ 0 & 0 & 0 & -b(1-a) \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \\ 0 & 0 & 0 & kb \end{array}\right).$$

When $b = 0$, there are infinitely many solutions, and when $b \neq 0$, there are no solutions.

We hence summarize our results as follows: (i) no solution: $a = 0$ and $b \neq 0$; $a = 2$ and $b \neq 0$; (ii) unique solution: $a \neq 0, 2$ and $b \in \mathbb{R}$; (iii) infinitely many solutions: $a = 0, 2$ and $b = 0$.

3. We have the following problem from Chapter 8 of *The Nine Chapters on the Mathematical Art:*

> If you sell two cows and five sheep and you buy 13 pigs, you gain 1,000 coins. If you sell three cows and three pigs and buy nine sheep, you break even. If you sell six sheep and eight pigs and you buy five cows, you lose 600 coins. What is the price of a cow, a sheep, and a pig?

Let x, y, and z be the prices, in coins, of cows, sheep, and pigs respectively. Then, we have the linear system

$$\begin{cases} 2x + 5y - 13z = 1000 \\ 3x - 9y + 3z = 0 \\ -5x + 6y + 8z = -600 \end{cases}.$$

We solve this linear system:

$$\left(\begin{array}{ccc|c} 2 & 5 & -13 & 1000 \\ 3 & -9 & 3 & 0 \\ -5 & 6 & 8 & -600 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{\frac{1}{3}R_1} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 2 & 5 & -13 & 1000 \\ -5 & 6 & 8 & -600 \end{array}\right) \cdots$$

$$\xrightarrow[R_3+5R_1]{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & 11 & -15 & 1000 \\ 0 & -9 & 13 & -600 \end{array}\right) \xrightarrow{R_3+\frac{9}{11}R_2} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & 11 & -15 & 1000 \\ 0 & 0 & \frac{8}{11} & \frac{2400}{11} \end{array}\right) \cdots$$

$$\xrightarrow{\frac{11}{8}R_3} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & 11 & -15 & 1000 \\ 0 & 0 & 1 & 300 \end{array}\right) \xrightarrow[R_2+15R_3]{R_1-R_3} \left(\begin{array}{ccc|c} 1 & -3 & 0 & -300 \\ 0 & 11 & 0 & 5500 \\ 0 & 0 & 1 & 300 \end{array}\right) \cdots$$

$$\xrightarrow{\frac{1}{11}R_2} \left(\begin{array}{ccc|c} 1 & -3 & 0 & -300 \\ 0 & 1 & 0 & 500 \\ 0 & 0 & 1 & 300 \end{array}\right) \xrightarrow{R_1+3R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1200 \\ 0 & 1 & 0 & 500 \\ 0 & 0 & 1 & 300 \end{array}\right).$$

Thus, a cow costs $x = 1200$ coins, a sheep costs $y = 500$ coins, and a pig costs $z = 300$ coins.

4. (a) Find a formula for the quadric surface $ax^2 + by^2 + cz^2 = d$ that passes through the points $(1,1,1)$, $(1,3,3)$, and $(-2,0,2)$.

   (b) The equation of a circle of radius r centered at the point (a,b) is given by $(x-a)^2 + (y-b)^2 = r^2$. Find the equation of the circle that passes through the points $(5,5)$, $(4,6)$, and $(6,2)$.

(a) Each of the given points must satisfy the given equation. In particular, substituting $(x,y,z) = (1,1,1)$, $(1,3,3)$ and $(-2,0,2)$ into the equation, we obtain a homogeneous system of linear equations of three equations in four unknowns a, b, c, d:

$$\begin{cases} a + b + c - d = 0 \\ a + 9b + 9c - d = 0 \\ 4a + 4c - d = 0 \end{cases}.$$

We solve the system by performing Gauss-Jordan elimination:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 1 & 9 & 9 & -1 & 0 \\ 4 & 0 & 4 & -1 & 0 \end{array}\right) \xrightarrow[R_3-4R_1]{R_2-R_1} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & 8 & 8 & 0 & 0 \\ 0 & -4 & 0 & 3 & 0 \end{array}\right) \cdots$$

$$\xrightarrow{\frac{1}{8}R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & -4 & 0 & 3 & 0 \end{array}\right) \xrightarrow[R_3+4R_2]{R_1-R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 4 & 3 & 0 \end{array}\right) \cdots$$

$$\xrightarrow{\frac{1}{4}R_3} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{4} & 0 \end{array}\right) \xrightarrow{R_2-R_3} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -\frac{3}{4} & 0 \\ 0 & 0 & 1 & \frac{3}{4} & 0 \end{array}\right).$$

The final augmented matrix corresponds to the linear system

$$\begin{cases} a - d = 0 \\ b - \frac{3}{4}d = 0 \\ c + \frac{3}{4}d = 0 \end{cases}.$$

To find a general solution, we assign an arbitrary parameter to d; thus, let $d = t$ for $t \in \mathbb{R}$. Then,

$$a = t, \quad b = \frac{3}{4}t, \quad c = -\frac{3}{4}t, \quad d = t, \quad t \in \mathbb{R}.$$

Note that there are infinitely many solutions, and any of the non-trivial solutions gives us a formula for the same quadric surface. Taking $t = 4$, for instance, we have

$$4x^2 + 3y^2 - 3z^2 = 4.$$

(b) Plugging each of the given points into the equation of the circle, we find that

$$\begin{cases} (5-a)^2+(5-b)^2 &= r^2 \\ (4-a)^2+(6-b)^2 &= r^2 \\ (6-a)^2+(2-b)^2 &= r^2 \end{cases}.$$

Expanding each of the expressions on the left-hand side, we obtain a system of nonlinear equations:

$$\begin{cases} a^2-10a+b^2-10b+50 &= r^2 \\ a^2-8a+b^2-12b+52 &= r^2 \\ a^2-12a+b^2-4b+40 &= r^2 \end{cases}.$$

We transform this into a linear system: let $x_1 = a^2$, $x_2 = a$, $x_3 = b^2$, $x_4 = b$, and $x_5 = r^2$. Then, we have the system

$$\begin{cases} x_1 - 10x_2 + x_3 - 10x_4 - x_5 = -50 \\ x_1 - 8x_2 + x_3 - 12x_4 - x_5 = -52 \\ x_1 - 12x_2 + x_3 - 4x_4 - x_5 = -40 \end{cases}.$$

We proceed with Gaussian elimination on the augmented matrix:

$$\left(\begin{array}{ccccc|c} 1 & -10 & 1 & -10 & -1 & -50 \\ 1 & -8 & 1 & -12 & -1 & -52 \\ 1 & -12 & 1 & -4 & -1 & -40 \end{array}\right) \xrightarrow[R_3-R_1]{R_2-R_1} \left(\begin{array}{ccccc|c} 1 & -10 & 1 & -10 & -1 & -50 \\ 0 & 2 & 0 & -2 & 0 & -2 \\ 0 & -2 & 0 & 6 & 0 & 10 \end{array}\right) \cdots$$

$$\xrightarrow{R_3+R_2} \left(\begin{array}{ccccc|c} 1 & -10 & 1 & -10 & -1 & -50 \\ 0 & 2 & 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 4 & 0 & 8 \end{array}\right) \xrightarrow[\frac{1}{4}R_3]{\frac{1}{2}R_2} \left(\begin{array}{ccccc|c} 1 & -10 & 1 & -10 & -1 & -50 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{array}\right) \cdots$$

$$\xrightarrow[R_2+R_3]{R_1+10R_3} \left(\begin{array}{ccccc|c} 1 & -10 & 1 & 0 & -1 & -30 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{array}\right) \xrightarrow{R_1+10R_2} \left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & -1 & -20 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{array}\right).$$

Letting $x_3 = s$ and $x_5 = t$, a general solution is given by

$$x_1 = -20 - s + t,\ x_2 = 1,\ x_3 = s,\ x_4 = 2,\ x_5 = t,\ s, t \in \mathbb{R}.$$

In terms of a and b, we thus require that

$$\begin{cases} x_1 = a^2 &= -20 - s + t, \\ x_2 = a &= 1, \\ x_3 = b^2 &= s, \\ x_4 = b &= 2, \\ x_5 = r^2 &= t, \quad s, t \in \mathbb{R}. \end{cases}$$

Hence, $a = 1$ and $b = 2$. To determine $r^2 = t$, we can use the expression for x_1:

$$r^2 = a^2 + b^2 + 20 = 1 + 4 + 20 = 25.$$

Thus, the equation of the circle containing the given points must be

$$(x-1)^2 + (y-2)^2 = 25.$$

5. Consider the standard basis vectors of $\mathbb{R}^3$, written as column matrices:

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

(a) Let $\boldsymbol{A}$ be an arbitrary 3×3 matrix. What are $\boldsymbol{A}\vec{e}_1$, $\boldsymbol{A}\vec{e}_2$, and $\boldsymbol{A}\vec{e}_3$?

(b) Let $\boldsymbol{B}$ be an arbitrary $n \times 3$ matrix, with columns $\vec{v}_1$, $\vec{v}_2$, and $\vec{v}_3$. What are $\boldsymbol{B}\vec{e}_1$, $\boldsymbol{B}\vec{e}_2$, and $\boldsymbol{B}\vec{e}_3$?

(c) In $\mathbb{R}^n$, we define the ith standard basis vector as the $n \times 1$ matrix:

$$\vec{e}_i = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \begin{array}{l} \\ \\ \\ \leftarrow i\text{th component} \\ \\ \\ \end{array}$$

If $\boldsymbol{A}$ is an $m \times n$ matrix, what is $\boldsymbol{A}\vec{e}_i$?

(d) Find a 3×3 matrix $\boldsymbol{A}$ such that

$$\mathbf{A}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{A}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{A}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 10 \\ 12 \end{pmatrix}.$$

(a) These are just the first, second, and third columns of $\boldsymbol{A}$.

(b) We have $\boldsymbol{B}\vec{e}_i = \vec{v}_i$ for $i = 1, 2, 3$.

(c) This is just the ith column of $\boldsymbol{A}$.

(d) Observe that the first two equations give us expressions for $\boldsymbol{A}\vec{e}_1$ and $\boldsymbol{A}\vec{e}_2$, which are precisely the first two columns of $\boldsymbol{A}$. To determine the third column of $\boldsymbol{A}$, observe that

$$\begin{aligned} \boldsymbol{A}\vec{e}_3 &= \boldsymbol{A}\left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \boldsymbol{A}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \boldsymbol{A}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 8 \\ 10 \\ 12 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}. \end{aligned}$$

Hence, $\boldsymbol{A}$ must be the matrix

$$\boldsymbol{A} = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}.$$

6. A linear system of the form $\boldsymbol{A}\vec{x} = \vec{0}$ is called *homogeneous*. Explain why each of the following statements are true.

   (a) All homogeneous systems are consistent.

   (b) A homogeneous system with fewer equations than unknowns has infinitely many solutions.

   (c) If $\vec{x}_1$ and $\vec{x}_2$ are solutions of the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$, then, $\vec{x}_1 + \vec{x}_2$ is a solution as well.

   (d) If $\vec{x}$ is a solution of the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$, then, $k\vec{x}$ is a solution as well for any constant $k \in \mathbb{R}$.

   (a) The zero vector $\vec{x} = \vec{0}$ is always a solution to any homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$; thus, all homogeneous systems are consistent. This solution is known as the *trivial solution*.

   (b) Since the system has fewer equations than unknowns, the size of $\boldsymbol{A}$ must be $m \times n$ with $m < n$: that is, $\boldsymbol{A}$ has more columns than rows. Thus, the reduced row-echelon form of $\left(\boldsymbol{A} \mid \vec{0}\right)$ must have at least one column that does not contain a leading entry. Since all homogeneous systems are consistent, we may assign an arbitrary parameter to the corresponding variable, and thus produce infinitely many solutions to the linear system.

   (c) Since $\vec{x}_1$ and $\vec{x}_2$ are solutions to $\boldsymbol{A}\vec{x} = \vec{0}$, we have $\boldsymbol{A}\vec{x}_1 = \boldsymbol{A}\vec{x}_2 = \vec{0}$. Observe, then, that

   $$\boldsymbol{A}\left(\vec{x}_1 + \vec{x}_2\right) = \boldsymbol{A}\vec{x}_1 + \boldsymbol{A}\vec{x}_2 = \vec{0} + \vec{0} = \vec{0},$$

   and $\vec{x}_1 + \vec{x}_2$ is a solution to $\boldsymbol{A}\vec{x} = \vec{0}$.

   (d) Observe that

   $$\boldsymbol{A}\left(k\vec{x}\right) = k\left(\boldsymbol{A}\vec{x}\right) = k\vec{0} = \vec{0},$$

   and so $k\vec{x}$ is a solution to $\boldsymbol{A}\vec{x} = \vec{0}$.

7. Suppose that the linear system $\boldsymbol{A}\vec{x} = \vec{b}$ has a solution given by $\vec{v}$.

   (a) Suppose that $\vec{x}_h$ is a solution to the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$. Show that $\vec{v} + \vec{x}_h$ is a solution of the system $\boldsymbol{A}\vec{x} = \vec{b}$.

   (b) If $\vec{w}$ is another solution of the system $\boldsymbol{A}\vec{x} = \vec{b}$, show that $\vec{w} - \vec{v}$ is a solution of the system $\boldsymbol{A}\vec{x} = \vec{0}$.

   (c) Suppose that $\boldsymbol{A}$ is a 2×2 matrix whose entries are not all zero. Furthermore, suppose that $\boldsymbol{A}\vec{x} = \vec{0}$ has at least one solution, $\vec{y} \neq \vec{0}$.

      i. Explain why the set of all solutions to $\boldsymbol{A}\vec{x} = \vec{0}$ forms a line through the origin in $\mathbb{R}^2$.

      ii. Describe the set of all solutions to $\boldsymbol{A}\vec{x} = \vec{b}$, where $\vec{b} \in \mathbb{R}^2$.

Note that since $\vec{v}$ is a solution of $\boldsymbol{A}\vec{x} = \vec{b}$, we have $\boldsymbol{A}\vec{v} = \vec{b}$.

(a) Since $\vec{x}_h$ is a solution to $\boldsymbol{A}\vec{x} = \vec{0}$, we have $\boldsymbol{A}\vec{x}_h = \vec{0}$. Observe that

$$\boldsymbol{A}(\vec{v} + \vec{x}_h) = \boldsymbol{A}\vec{v} + \boldsymbol{A}\vec{x}_h = \vec{b} + \vec{0} = \vec{b},$$

and $\vec{v} + \vec{x}_h$ is a solution of $\boldsymbol{A}\vec{x} = \vec{b}$.

(b) Since $\vec{w}$ is a solution to $\boldsymbol{A}\vec{x} = \vec{b}$, we have $\boldsymbol{A}\vec{w} = \vec{b}$. Observe that

$$\boldsymbol{A}(\vec{w} - \vec{v}) = \boldsymbol{A}\vec{w} - \boldsymbol{A}\vec{v} = \vec{b} - \vec{b} = \vec{0},$$

and so $\vec{w} - \vec{v}$ is a solution of $\boldsymbol{A}\vec{x} = \vec{0}$.

(c) i. Let $\boldsymbol{R}$ denote the reduced row-echelon form of $\boldsymbol{A}$. Since $\boldsymbol{A}\vec{x} = \vec{0}$ has at least one non-trivial solution $\vec{y} \neq \vec{0}$ (in addition to the trivial solution $\vec{x} = \vec{0}$), $\boldsymbol{A}\vec{x} = \vec{0}$ must have infinitely many solutions, and so $\boldsymbol{R}$ must have at least one column without a leading entry. Since $\boldsymbol{A}$ is not the zero matrix, one of the columns of $\boldsymbol{R}$ must contain a leading entry. In particular, $\boldsymbol{R}$ has one column without a leading entry, and one column with a leading entry. Thus, the general solution to $\boldsymbol{A}\vec{x} = \vec{0}$ involves one arbitrary parameter: say

$$\vec{x} = \vec{p} + t\vec{w} \text{ for some } \vec{p}, \vec{w} \in \mathbb{R}^2.$$

Since the zero vector $\vec{x} = \vec{0}$ is a solution to $\boldsymbol{A}\vec{x} = \vec{0}$, we may take $\vec{p} = \vec{0}$. Thus, the solutions to $\boldsymbol{A}\vec{x} = \vec{0}$ form a line through the origin in $\mathbb{R}^2$.

ii. Suppose that $\vec{b} \neq \vec{0}$ in $\mathbb{R}^2$. Since $\vec{v} + \vec{x}_h$ is a solution to $\boldsymbol{A}\vec{x} = \vec{b}$, we may express the solutions to $\boldsymbol{A}\vec{x} = \vec{b}$ as:

$$\vec{x} = \vec{v} + \vec{x}_h = \vec{v} + t\vec{w}.$$

This is a line parallel to the line containing the solutions of $\boldsymbol{A}\vec{x} = \vec{0}$: in particular, it is the line of solutions to $\boldsymbol{A}\vec{x} = \vec{0}$, translated by $\vec{v}$.